

LAB # 11

Construct java programs using the concepts of method and constructor overloading

Objective

Implementation of overloading methods and overloading constructors.

Lab Goals

- Understand the concept of constructors overloading.
- Understand the concept of methods overloading.

Theory

Method overloading

Method overloading is a concept that allows declaring multiple methods with same name but different parameters in the same class. Method overloading is one of the ways through which java supports polymorphism. Polymorphism is a concept of object oriented programming that deals with multiple forms. We will cover polymorphism in separate topics later on.

Method overloading can be done by changing number of arguments or by changing the data type of arguments. If two or more method have same name and same parameter list but differs in return type can not be overloaded.

Ways of method overloading.

1. Different data type of argument
2. Different number of arguments.

Method overloading by changing data type of arguments.**Example**

```
class Calculate
{
    void sum (int a, int b)
    {
        System.out.println("sum is" +(a+b)) ;
    }
    void sum (float a, float b)
    {
        System.out.println("sum is" +(a+b));
    }
    Public static void main (String[] args)
    {
        Calculate cal = new Calculate();
        cal.sum (8,5);    //sum(int a, int b) is method is called.
        cal.sum (4.6f, 3.8f); //sum(float a, float b) is called.
    }
}
```

Output:**Sum is 13****Sum is 8.4**

Method overloading by changing no. of argument.**Example**

```
class Demo
{
    void multiply(int l, int b)
    {
        System.out.println("Result is" + (l * b));
    }
    void multiply(int l, int b, int h)
    {
        System.out.println("Result is" + (l * b * h));
    }
    public static void main(String[] args)
    {
        Demo ar = new Demo();
        ar.multiply(8, 5); // multiply(int l, int b) is method is called
        ar.multiply(4, 6, 2); // multiply(int l, int b, int h) is called
    }
}
```

Output:

```
Result is 40
Result is 48
```

Constructor overloading

Like methods, constructors can also be overloaded. Constructor overloading is a concept of having more than one constructor with different parameters list, in such a way so that each constructor performs a different task.

Example:

```
//Java program to overload constructors
class Student5{
    int id;
    String name;
    int age;
    //creating two arg constructor
    Student5(int i,String n){
        id = i;
        name = n;
    }
    //creating three arg constructor
    Student5(int i,String n,int a){
        id = i;
        name = n;
        age=a;
    }
    void display(){System.out.println(id+" "+name+" "+age);}

    public static void main(String args[]){
        Student5 s1 = new Student5(111,"Kiran");
        Student5 s2 = new Student5(222,"Ayan",25);
        s1.display();
        s2.display();
    }
}
```

Output:

```
111 Kiran 0
222 Ayan 25
```

Exercise

- A. Create a class to print the area of a square and a rectangle. The class has two methods with the same name but different number of parameters. The method for printing area of rectangle has two parameters which are length and breadth respectively while the other method for printing area of square has one parameter which is side of square.
- B. Define a class with the name "Mathematics" and define a method "Square()" for calculating the square of any given number. Now redefine this method with different type of argument variables, but with the same name "Square()".
- C. Write a program to print the names of students by creating a Student class. If no name is passed while creating an object of Student class, then the name should be "Unknown", otherwise the name should be equal to the String value passed while creating object of Student class.